Gondola feasibility check V2

Four research agents checked integrations, compliance, ML and competition. We built a working test repo, generated a year of demo data with 2,355 planted problems, scored every agent against that answer key, and had an independent QA agent try to break it. This is what holds up, where it fails, and what we changed.

Verdict

- **Buildable and demo-ready.** The full close runs in about 7 seconds (2 seconds once warm), finished all 47 nights of a replay, and survives any single agent failing.
- **Same-night reconciliation of delivery-app money is not possible** with real data. DoorDash's reporting data arrives the next evening and Uber's can take up to 72 hours. Gondola now estimates tonight and verifies each app when its report lands.
- **Gondola can't file disputes or log into the Merchant Portal.** DoorDash's terms forbid both, so Books produces a checklist the manager submits.
- **The biggest risk is the market, not the tech.** Square's free Managerbot covers about 6 of the 9 agents. Lead with what nobody ships: forecast-timed, per-channel sold-out pausing, plus cash and three-app reconciliation at close.

94%
of store-caused DoorDash cancellations Guard would have prevented (1,432 of 1,522)

24 → 8
chashu sell-out days in 90, with Gondola's prep plan vs the manager's habit

95% / 88%
precision / recall on disputable error charges (realistic noise added)

8.1%
daily forecast error over 120 days, vs 9.1% for same-weekday averages

V2 re-run

V2 keeps the nightly close-out pipeline and applies the corrections: payouts verified as reports land, a dispute checklist, portion counts at close, tomorrow's card (prep plus pause time per app), Guard with a recount, and one bundled text. We re-ran every test, and a second independent QA agent re-checked the 23 earlier issues and attacked the new parts.

Check	Result
Tests and demo	31 of 31 tests pass. Demo close in 1.8 s warm (7.7 s cold), with the same text: cash $23.50 short, $86.40 recoverable on Wednesday's DoorDash report, chashu sold out at 8:39 PM and live 45 minutes (12 cancels).
47-night replay	47 of 47 nights done, median 2.2 s, texts on 14 nights, 0 cross-check mismatches. 30 PDFs stored and 17 expired; 2 monthly summaries; a 40-day-old report rebuilt.

Guard under real error (90 days, status quo 294 store-caused app cancellations)	Recount when Gondola *believes* about 120 portions are left (about 0.8 prompts a night): **0–6** cancellations at 5–10% count error with a pause delay of up to 5 minutes, and 19–23 at 15 minutes. Without the recount, a 5% error gives 192–402. The recount is required.
Earlier issues	18 of 23 fixed, 4 partly, 1 open (Friday over-portioning). None regressed.
New issues found and fixed	The first recount trigger read the kitchen's true stock, which a real system can't know. It now uses only the believed count and the result above is honest. Also fixed: Christmas forecast for a new store (holiday prior now in the fallback forecast; closed days prep nothing); bad leftover counts (warnings, capped at target); portions per order now learned from days that didn't sell out; the demo script no longer deletes eval results.
Still caveated	The +$11.9k prep result uses true demand, including sales lost after a sell-out, so it is an upper bound. All data is synthetic. Texts and the LLM wording need wiring (Twilio paid account, ADK key) before the demo.

What we built and tested

The repo `Gondola/gondola-test` holds a working version of the close pipeline. The connectors read files shaped like Square orders, cash drawer shifts and the apps' payout exports. The ten agents' tools are real code: reconciliation, ingredient ledger, analysis, LightGBM forecasting with conformal ranges, newsvendor prep and an OR-Tools prep schedule, sold-out guard, SMS and email rules, a 2-page PDF report, a 30-day report bucket, and the Ask router. The Google ADK graph builds, with a guardrail callback on every agent. Running it with a live model only needs an API key.

Demo data (fictional Hanabi Ramen, Squirrel Hill)	Contents
365 days, Nov 6 2025 – Nov 5 2026	171,929 orders, 385,019 order lines and 118,599 payout lines across in-store, DoorDash, Uber Eats and Grubhub. Also weather, Penguins, Steelers and Pirates home games, holidays, CMU/Pitt semesters, labour, invoices, waste and weekly counts.
Answer key (never read by the pipeline)	2,355 planted problems: disputable error charges, orders missing from payouts, commission overcharged on 5 days, 11 cash shortages, beer shrink in March, Friday noodle over-portioning, over-scheduled days. Also true inventory, lost sales during sell-outs, and tomorrow's actual orders.
Demo night, Thu Nov 5	Rain and a Penguins game. Chashu sells out at 8:39 PM and stays live on the apps for 45 minutes (12 cancels). The cash drawer is $23.50 short, and Wednesday's DoorDash report arrives with $86.40 in charges that look wrong.

The manager's text on demo night: "Gondola close Thu Nov 5: 3 things need you. 1) Cash drawer $23.50 short 2) DoorDash Wed: $86.40 looks recoverable (3 charges to dispute) 3) Chashu sold out 8:39 PM, still live on apps 45 min (12 cancels)..."

Results by agent

Agent	What held up	Where it falls short

Books	Missing orders 289/289, commission-overcharge days 5/5, cash shortages 11/11. Disputable error charges: 95% precision, 88% recall. Flagged $20,102 of $21,296 truly recoverable over the year.	347 partial charges can't be judged from order data, so they go to "review" (47 of them were truly wrong). Cash flags at the $5 level have 8 false alarms, none of them texted.
Stock	Beer shrink flagged in 3 of 4 weeks, with no false flags on other ingredients.	**Missed the 12% Friday noodle over-portioning in all 52 weeks:** weekly counts dilute it to about 2%. This needs daily spot counts on 2–3 high-volume items.
Demand	8.1% average daily error over 120 days (naive 9.1%). Tomorrow: 590 forecast vs 607 actual, inside the 532–678 range. 80% ranges covered 88% of days. Hourly shape error 11%.	Only about 11% better than a simple same-weekday average. 15-minute shape error is 19%. A single year can't teach holidays (Labor Day was missed by 60%), so holiday priors are needed. With perfect weather the error would drop to 6.9%, so weather-forecast quality matters.
Plan	Over 90 days, sell-out days fall from 24 to 8 and lost margin from $16,434 to $3,009, for $1,262 more in leftovers. The CP-SAT prep schedule always solves.	Keeping the leftovers usable depends on a 3-day hold for chashu; it's more leftover, sold the next day.
Guard	Replayed over all 364 days, it would have prevented 1,432 of 1,522 store-caused DoorDash cancellations.	It holds back 226 app orders ($8,098) for dine-in. That's the right trade for DoorDash's score, but the manager has to agree to it.
Sentinel	One bundled text per close, at most 320 characters, with STOP, quiet hours, opt-in and dedupe. Texts went out on 14 of 47 nights, about twice a week.	**The first rules texted on 37 of 47 nights.** Alert fatigue is a real failure mode, and the thresholds need tuning per store.
Report	A 2-page PDF in about 1 second. Stored 30 days, then expired; any day can be rebuilt from its findings file. Monthly summaries were written for September and October.	This cut the 4-page report to 2 pages. Two pages is still more than a manager reads at 11 pm; the text is the main output.
Ask	25 of 25 on the scored set. On QA's 62 hard questions: no crashes, 51 answered, and 11 clarifying questions instead of guesses.	The offline router is brittle by design. The live version needs the LLM plus the same typed tools, behind the guardrail.
Coordinator	Median 2.3 s per close in the replay, 7 s cold; 0 cross-check mismatches. If any one agent fails, the close still finishes and a failure text goes out. A second tap is idempotent.	On the first night, a two-week backlog of payout reports lands at once (handled: it goes in the report, not the text).

Where it fails in the real world, ranked

	Risk	What we found	What to do
1	**Same-night app reconciliation**	DoorDash's Reporting API refreshes once a day at 7 PM PST with the previous day, and needs early-access approval. Uber can take up to 72 hours. No public Grubhub payout API. The POS doesn't see DoorDash fees.	Built: "estimated tonight, verified when the report lands". Ask the DoorDash sponsor for a sample Financial export and Reporting API access.

2	**Free platforms already do most of it**	Square Managerbot (free): recaps, shortage flags, availability changes, schedules. DoorDash Assistant: reports, out-of-stock, error-charge advice. Toast IQ: Q&A and 86-ing. Loop, Voosh and Otter recover disputes. Tenzo sends an end-of-day report with a forecast. Marble (YC S26) targets the whole back of house.	Pitch three outcomes ("money found", "what runs out", "what to prep"). Make forecast-timed per-channel pausing the headline, and position it as a DoorDash extension that cuts cancellations.
3	**DoorDash terms**	No third-party Portal access or dispute filing. The dispute form is tick-boxes, needs an Admin or Store Manager, and has a 14-day window.	Built: a checklist with days left. Never hold portal credentials. DoorDash data comes from merchant-uploaded CSVs or official APIs.
4	**Sold-out writes**	Toast's standard API is read-only. Square's only way to mark an item sold out is to set its inventory count to 0. Middleware item APIs are gated.	Demo on a Square sandbox. Say clearly that Toast needs partner access.
5	**SMS at the hackathon**	A Twilio trial can't send custom message bodies. US 10DLC vetting takes up to 5 business days and toll-free about 3–5.	Use a paid account with a Canadian number to the team's phones, and test a day early. Build an on-screen SMS view as a fallback. Start toll-free verification now.
6	**Trust in numbers**	QA showed a number-only guardrail can't tell which app or direction a number refers to.	Built: templates own every money sentence, the LLM may only rephrase, and every SMS and report is checked before sending.
7	**Ingredient data**	Independents rarely keep recipes and counts. Weekly counts missed the Friday over-portioning.	Start in "forecast mode" (no recipes needed). Add daily spot counts for 2–3 items later.
8	**Forecast credibility**	There's no real store data, and the gain over a naive baseline is modest. Chronos-2 sees only about 85 days of 15-minute history. MAPIE's time-series methods give no coverage guarantee. Gemini 2.5 Flash is being retired.	Show the backtest against the naive baseline honestly. Forecast daily totals and split by similar-day shape (built). Pin a current model.
9	**Record keeping**	The IRS expects business records kept 3–4 years, and the 30-day PDF expiry could collide with that.	Label the emailed PDF as the merchant's copy, keep cash counts with the 365-day metrics, and offer an export.
10	**Synthetic data**	The detection rules and the generator share assumptions, so these scores are an upper bound.	Say so on the slide. Re-score on a real Report Builder export as soon as one is available.

What the independent QA pass broke, and what changed

A separate agent that hadn't seen the code found 23 issues: 3 critical, 5 high, 11 medium, 4 low. We fixed 19, partly fixed 3 and documented 1. All 26 tests passed in the first pass (31 in V2), and its reproduction cases now behave.

Issue	Severity	Fix
Payouts marked "verified" before the app's report arrived	Critical	Availability and verification are now tracked per day *and* app. Uber Eats for Wednesday now shows "pending" on Thursday night.
Running the demo twice lost the headline and sent a second text	Critical	A re-run redoes its own checks. One text per close date, however many taps (3 runs give the same 30 findings and 1 text).
Ask gave confident wrong answers ("orders tomorrow" returned today's)	Critical	New router with forecast intent, explicit years, ordinals, holidays, item names, channel vs channel and typo fixing. It asks when unsure and never defaults.
Closing past dates crashed the forecast; a cache bug changed the demo's numbers	High	Past closes rebuild the database as it was that night. The cache is never overwritten with a filtered copy.
The guardrail was never called, and passed IDs, dates and near-misses	High	Rewritten: numeric fields only, matched at the precision written. Wired into the SMS, the report and every ADK agent.
Closed day (Christmas) and impossible dates crashed	High	Closed-day path with a short report and no failure text. Ask explains the closure and rejects "Feb 30" politely.
The demo's sell-out never surfaced; the cancel alert was mislabelled and noisy	Medium	New "sold out tonight" finding with minutes live on the apps. The alert is renamed "store-caused cancellations" and only texts when it crosses 1.1%.
Duplicate lines, reversed charges, open or multiple cash drawers	Medium	Duplicate lines removed on load, overpayments flagged, reversed charges dropped, all drawers summed, "drawer not counted" reported.
Holidays, school breaks and nan values in reports; data leakage in raw files	Medium	Holiday and CMU/Pitt break context, "no baseline yet", future rows removed, timezone-correct timestamps.
Circular scores; noodle over-portioning missed	Partly / open	Partial refunds and misspecified holiday priors added. Over-portioning is documented and needs daily spot counts.

What to change before the hackathon

Pitch

- Three outcomes, not nine agents: money found tonight, what runs out tomorrow, what to prep.
- Headline Guard: "pause chashu on the apps at 8:25, keep it for dine-in". Back it with the 94% replay result.

Build

- Twilio paid account and Canadian number; test texts a day early; on-screen SMS fallback.
- Square sandbox for orders and the sold-out write; DoorDash data from a CSV in the documented Reporting API format.

- Answer "why wouldn't DoorDash build this" up front: it reduces DoorDash's own cancellations. Partner or acquire.
- Show the forecast backtest against the naive baseline, and say the data is synthetic.
- Drop "one integration provider per store" from the deck: no DoorDash document states it.

- Pin a current Gemini or Claude model, paid tier only (free-tier inputs may be reviewed by people).
- Cut video counting from the demo; at most, a photo with a confirm button.
- Price at $49–79 a month, or a share of recovered money for Books.

How to run it

`python scripts/run_demo.py` runs the demo night end to end. `python -m pytest -q tests` runs 26 checks. `python scripts/evaluate.py` re-scores everything (about 6 minutes) into `out/eval.json`. The research notes from the four agents and the QA report are in `docs/feasibility`.

Sources

DoorDash Reporting API help (7 PM PST refresh): https://help.doordash.com/en-ca/merchants/article/how-can-i-access-the-doordash-reporting-api

DoorDash Reporting API reports: https://developer.doordash.com/docs/reporting/overview/available_reports

DoorDash Reporting API access: https://developer.doordash.com/docs/reporting/how_to/get_access

DoorDash error charges and third-party terms: https://help.doordash.com/en-us/merchants/article/what-are-order-error-adjustments

DoorDash merchant security terms: https://help.doordash.com/en-us/legal/article/mx-security-terms

DoorDash Assistant: https://help.doordash.com/en-us/merchants/article/doordash-assistant

DoorDash Marketplace item status: https://developer.doordash.com/docs/marketplace/how_to/item_status

Uber Eats reporting: https://developer.uber.com/docs/eats/guides/reporting

Grubhub developer features: https://grubhub-developers.zendesk.com/hc/en-us/articles/115002660546-Feature-Overview

Square Cash Drawer reporting: https://developer.squareup.com/docs/cashdrawershift-api/reporting

Square: no cash drawer webhook: https://developer.squareup.com/forums/t/feature-request-add-a-webhook-for-opening-and-closing-a-cash-drawer/19098

Square: DoorDash fees don't sync: https://community.squareup.com/t5/Orders-Menu-Items-Catalog/Doordash-sync-fees-to-Square-for-accurate-reports/td-p/804744

Square: DoorDash orders as pickup: https://developer.squareup.com/forums/t/new-doordash-integration-data/23865

Square sold-out status: https://developer.squareup.com/docs/inventory-api/monitor-sold-out-status-on-item-variation

Square Managerbot: https://squareup.com/us/en/press/managerbot-open-beta

Toast API access requirements: https://doc.toasttab.com/doc/devguide/devApiAccessRequirements.html

Toast standard access scopes: https://doc.toasttab.com/doc/devguide/devApiAccessScopes.html

Toast IQ: https://pos.toasttab.com/news/toast-expands-toast-iq-smart-ai-assistant

Toast stock depletion (manual 86): https://intercom.help/toast-central/en/articles/13793466-toast-inventory-stock-depletion-canada-ireland-and-uk

Deliverect reporting webhooks: https://developers.deliverect.com/reference/reporting-webhooks

Deliverect AI agents: https://restauranttechnologynews.com/2026/04/deliverect-launches-ai-agents-and-smart-assistants-to-boost-restaurant-sales-and-protect-digital-revenue/

Loop Series A: https://ascendants.in/business-stories/loop-ai-raises-14m-series-a-restaurant-back-office-ai/attachment/loop-ai/

Voosh Dispute Manager: https://www.fsrmagazine.com/content/voosh-ai-launches-dispute-manager-solution-restaurant-profit-protection

Tenzo end-of-day report: https://www.gotenzo.com/resources/insight/what-does-the-ideal-end-of-day-report-look-like-for-hospitality-businesses

Marble (YC S26): https://runtimewire.com/article/marble-ai-restaurant-back-office-yc-s26

Restaurant tech spend: https://www.fsrmagazine.com/feature/restaurants-reach-a-technology-turning-point-rooted-in-simplicity

MarginEdge pricing: https://www.marginedge.com/pricing/

Twilio trial limits: https://www.twilio.com/docs/messaging/guides/how-to-use-your-free-trial-account

Twilio A2P 10DLC: https://www.twilio.com/docs/messaging/compliance/a2p-10dlc

Twilio 10DLC vetting: https://support.twilio.com/hc/en-us/articles/11587910480155-A2P-10DLC-Campaign-Vetting-FAQ

Twilio Canada SMS: https://twilio.com/guidelines/ca/sms

Twilio opt-out keywords: https://support.twilio.com/hc/en-us/articles/223134027-Twilio-Support-for-Opt-Out-Keywords-SMS-STOP-Filtering

SendGrid trial: https://support.sendgrid.com/hc/en-us/articles/35270136965403

18 Pa.C.S. 5702 definitions: https://law.justia.com/codes/pennsylvania/2010/title-18/chapter-57/5702/

IRS record keeping: https://irs.gov/businesses/small-businesses-self-employed/how-long-should-i-keep-records

Gemini API terms: https://ai.google.dev/gemini/terms

Chronos-2: https://www.amazon.science/blog/introducing-chronos-2-from-univariate-to-universal-forecasting

Chronos-2 model card: https://huggingface.co/amazon/chronos-2/blob/main/README.md

Decathlon on Chronos-2: https://aws.amazon.com/blogs/machine-learning/how-decathlon-runs-demand-forecasting-at-scale-with-chronos-2/

Restaurant forecasting accuracy research: https://www.crunchtime.com/blog/new-research-forecasting-accuracy-averages-just-60-even-though-most-restaurants-use-tech

MAPIE time-series regressor: https://mapie.readthedocs.io/en/stable/generated/mapie.regression.TimeSeriesRegressor.html

ADK parallel agents: https://adk.dev/agents/workflow-agents/parallel-agents/

Gemini 2.5 shutdown: https://tokencost.app/blog/gemini-2-5-shutdown-october-20-migration-cost